Clustering

In [ ]:
```
from sklearn import datasets
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
from sklearn.cluster import KMeans
```

In [ ]:
```
df = datasets.load_iris()
y = df.target
yframe=pd.DataFrame(y)
dfr = pd.DataFrame(data=df.data,
                  columns=df.feature_names)
dfr["target"] = yframe
dfr
```

Out[ ]:

	sepal length (cm)	sepal width (cm)	petal length (cm)	petal width (cm)	target
0	5.1	3.5	1.4	0.2	0
1	4.9	3.0	1.4	0.2	0
2	4.7	3.2	1.3	0.2	0
3	4.6	3.1	1.5	0.2	0
4	5.0	3.6	1.4	0.2	0
...	...	...	...	...	...
145	6.7	3.0	5.2	2.3	2
146	6.3	2.5	5.0	1.9	2
147	6.5	3.0	5.2	2.0	2
148	6.2	3.4	5.4	2.3	2
149	5.9	3.0	5.1	1.8	2

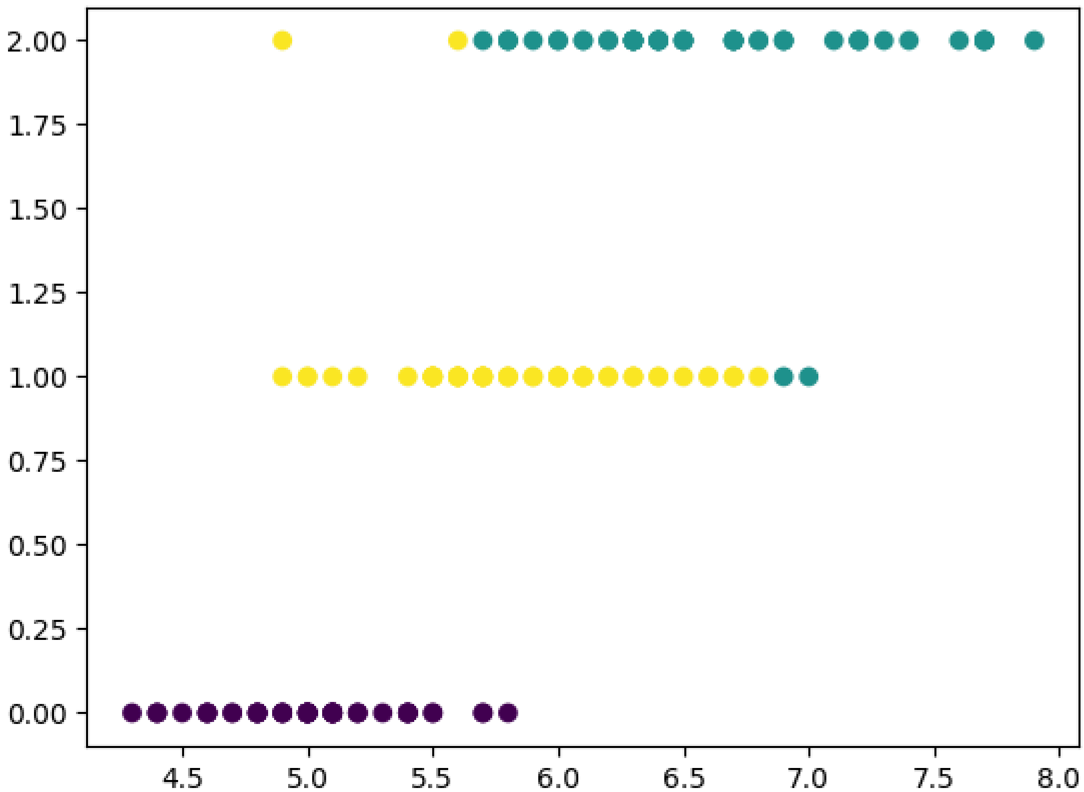
150 rows × 5 columns

In [ ]:
```
kmeans_model =KMeans(n_clusters=3)
y_kmeans = kmeans_model.fit(dfr[['sepal length (cm)','target']])
dfr['kmeans_3']=kmeans_model.labels_
dfr

plt.scatter(x=dfr['sepal length (cm)'],y=dfr['target'],c = dfr['kmeans_3'])
```

```
/usr/local/lib/python3.9/dist-packages/sklearn/cluster/_kmeans.py:870: FutureWa
rning: The default value of `n_init` will change from 10 to 'auto' in 1.4. Set
the value of `n_init` explicitly to suppress the warning
  warnings.warn(
```

Out[ ]: <matplotlib.collections.PathCollection at 0x7f0dfdeec790>



In [ ]:
```
#Intrinsic Method
from sklearn.metrics import silhouette_score
silhouette_score(dfr[['sepal length (cm)','target']], dfr['kmeans_3'], metric
```

Out[ ]: 0.560160227656915

In [ ]:
```
#Adjusted Rand Index
from sklearn.metrics.cluster import adjusted_rand_score
adjusted_rand_score(df['target'], dfr['kmeans_3'])
```

Out[ ]: 0.9221551020408163

In [ ]:
```
#Mutual Information
from sklearn.metrics.cluster import normalized_mutual_info_score
normalized_mutual_info_score(df['target'], dfr['kmeans_3'])
```

Out[ ]: 0.8980870992332958